

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV **COURSE CODE:** DSA02

COURSE OUTCOME COVERED

CO4: Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)

Assessment Tool Weightage: 20%

QUESTIONS: 2 Total Marks:20 Annexure A

Comparative Analysis

Code of Conduct:

*I JAASIEL J S (192524046) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies	
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive	
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided	

Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated	
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand	

Annexure A

Comparative Analysis

1. CNNs vs. Vision Transformers (ViTs) for Object Detection Scenario: A self-driving car company needs to detect rare urban obstacles (traffic cones, animals) in real-time using limited labeled datasets.

Questions:

- Compare CNNs and Vision Transformers in terms of:
 - Feature extraction capabilities for local vs. global context
 - Performance with small or sparse objects
 - Computational complexity and inference speed for embedded systems

ANS:

Comparison of CNNs and Vision Transformers

Parameter	CNNs	Vision Transformers (ViTs)
Feature extraction	CNNs are strong at extracting local features such as edges, textures, and shapes using convolutional filters.	ViTs use self-attention to model relationships between distant regions and capture more global context.
Small or sparse objects	CNNs can detect small objects effectively when appropriate feature pyramids and high-	ViTs can benefit from global context, which can help when small or sparse obstacles need

Parameter	CNNs	Vision Transformers (ViTs)
	resolution feature maps are used.	information from surrounding regions.
Computational complexity	Generally lower and more efficient for deployment, especially with lightweight CNN architectures.	Self-attention can require more computation and memory, particularly for high-resolution images.
Inference speed	Usually faster and better suited to real-time embedded inference.	Often slower and more resource-intensive, especially on constrained hardware.
Embedded systems	More suitable because lightweight CNNs can be optimized for limited memory and compute.	More difficult to deploy efficiently when computational resources are limited

2. Discuss which method is better suited for real-time deployment and how transfer learning or data augmentation could impact performance.

ANS:

1) For a self-driving car with **limited computational resources and a real-time requirement**, **CNNs are generally better suited for deployment**. Their efficient local feature extraction and availability of lightweight architectures make them suitable for embedded systems.

ViTs can provide stronger global-context modeling, but their higher computational and memory requirements can make real-time deployment more challenging on resource-constrained hardware.

2) Transfer learning can improve performance when the labeled dataset is small. A CNN or ViT pretrained on a large image dataset can be fine-tuned using the limited urban-obstacle dataset, reducing the amount of training data required and improving feature learning.

Data augmentation can increase the effective training-data diversity by

generating variations such as changes in scale, position, brightness, rotation, and cropping. This can improve robustness to rare obstacles and different road conditions.

2. PCA-Based Shape Priors vs. Graph-Based Learning for 3D Shape Correspondence

Scenario: An animation studio is aligning 3D human face meshes for morphing applications. Meshes vary in topology and vertex density.

Questions:

1. Compare PCA-based shape priors and graph-based learning in terms of:
 - Ability to handle non-rigid deformations
 - Robustness to varying mesh resolutions and noise
 - Ease of implementation and computational cost

ANS:

Parameter	PCA-Based Shape Priors	Graph-Based Learning
Non-rigid deformations	Works well for deformations that follow the learned principal modes, but is limited to the variations present in the training data.	Better suited to complex and non-rigid deformations because it can learn relationships between connected mesh structures.
Varying mesh resolutions and noise	Less robust when meshes have different vertex densities or significant noise; usually requires alignment and normalization.	More robust to varying mesh structures and can learn local and global relationships from graph connectivity.
Implementation and computational cost	Simpler to implement and generally less computationally expensive.	More complex to implement and usually requires greater computational resources and training.

2. Analyze which approach would provide more accurate correspondences for realistic morphing and why.

ANS: Graph-based learning is generally more suitable for realistic 3D face morphing. It can model complex non-rigid deformations and relationships between neighboring mesh vertices, making it more adaptable to differences in topology, vertex density, and facial shape.

PCA-based shape priors are simpler and computationally cheaper, but their correspondence quality is limited by the principal shape variations learned from the training data.

Therefore, for accurate correspondence in realistic morphing applications, graph-based learning is preferable, while PCA is more suitable when the meshes are well aligned and their variations are relatively limited.